

# Supplementary Materials

## Continuous Grip-Force Dynamics as an Online Biophysical Signal of Perceptual Strain

Mohammad Dastgheib

2026-06-24

### Table of contents

|          |                                                 |          |
|----------|-------------------------------------------------|----------|
| <b>1</b> | <b>Signal processing and feature extraction</b> | <b>1</b> |
| <b>2</b> | <b>Model specifications</b>                     | <b>1</b> |
| <b>3</b> | <b>Sample coverage and attrition</b>            | <b>2</b> |
| <b>4</b> | <b>Additional figures and tables</b>            | <b>3</b> |
| <b>5</b> | <b>Reproducibility</b>                          | <b>4</b> |

Main manuscript: [manuscript.qmd](#) (HTML/PDF) · [manuscript-biorxiv.qmd](#) (bioRxiv preprint PDF)

## 1 Signal processing and feature extraction

Grip-force preprocessing followed the pipeline documented in the main Methods and in `analysis/01_ingest.qmd`:

1. Target-normalized error:  $e(t) = (F(t) - F_{\text{target}})/F_{\text{target}}$
2. Linear detrending within each trial epoch
3. Fourth-order zero-phase Butterworth high-pass filter at 0.5 Hz (`sosfiltfilt`)
4. Welch PSD estimation (Hann window, 50% overlap, ~0.5 Hz resolution)
5. Band integration: visuomotor tracking (0.5–3.0 Hz) and physiological tremor (8.0–12.0 Hz)

Epochs spanned grip-cue onset through 1 s after probe onset. All spectral features were log-transformed before modeling.

## 2 Model specifications

**Mixed-effects logistic regression.** Perceptual accuracy was modeled as:

$$\text{logit } P(\text{correct}) = \beta_0 + \beta_1 \text{modality} + \sum_{k=1}^4 \beta_k \text{poly}_k(\text{difficulty}) + \beta_5 z(\text{tracking}) + \beta_6 z(\text{tremor}) + u_i$$

where  $u_i \sim \mathcal{N}(0, \sigma^2)$  is a random intercept for participant  $i$ , and  $z(\cdot)$  denotes within-condition standardized band power. Primary inference used high-grip trials only (40% MVC); low-grip models used the same structure.

**Cross-validated machine learning.** Random-forest and logistic classifiers were trained with subject-wise cross-validation (leave-one-subject-out) on high-grip trials. Feature sets compared:

- **Baseline:** stimulus difficulty level, task modality
- **DSP:** baseline features plus tracking- and tremor-band power

Performance was evaluated using area under the ROC curve (AUC).

Table 1: Cross-validated AUC for perceptual accuracy prediction (high-grip trials).

| Feature set | Model        | AUC    | N trials |
|-------------|--------------|--------|----------|
| Baseline    | Logistic     | 0.6170 | 7001     |
| Baseline    | RandomForest | 0.8498 | 7001     |
| DSP         | Logistic     | 0.6279 | 6911     |
| DSP         | RandomForest | 0.8553 | 6911     |

## 3 Sample coverage and attrition

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

| Stage                                     | Subject-<br>tasks | Tri-<br>als | %<br>retain-<br>able | Note                                     |
|-------------------------------------------|-------------------|-------------|----------------------|------------------------------------------|
| Oddball subject-tasks<br>in behavioral v3 | 136               | 18,099      | —                    | aud + vis; runs merged into trial counts |
| Excluded: aborted<br>mid-task             | 13                | 478         | —                    | beh_trials < 100                         |

Table 2: Grip-force data attrition funnel from behavioral oddball tasks to analysis-ready trials.

| Stage                                        | Subject-<br>tasks | Tri-<br>als | %<br>retain-<br>able | Note                                                                             |
|----------------------------------------------|-------------------|-------------|----------------------|----------------------------------------------------------------------------------|
| Retainable subject-tasks                     | 123               | 17,621      | 100.0%               | individual missed trials already removed in v3                                   |
| No grip files (entire subject $\times$ task) | 1                 | 128         | 0.7%                 | zero grip CSVs found; all behavioral trials in block excluded                    |
| Partial grip (behavioral > on-disk grip)     | 36                | 3,678       | 20.9%                | some grip files exist but fewer trials than behavioral (often 1 run = 30 trials) |
| Grip trials on disk                          | 122               | 14,430      | 81.9%                | fatigue-break runs merged per session; retainable subject-tasks only             |
| Feature extraction loss                      | 95                | 671         | 3.8%                 | grip trials present but spectral features not extracted (short epochs, etc.)     |
| Analysis set (trial features extracted)      | 122               | 13,759      | 78.1%                | primary grip force modeling cohort                                               |

## 4 Additional figures and tables

**Single-trial signal chain.** Representative high-grip incorrect trial showing the transformation from raw force to normalized error, rolling variance, and Welch PSD with analysis bands marked.

**Probe-locked spectrogram.** Time–frequency power aligned to probe onset ( $t = 0$ ), with normalized error in the upper panel.

**Population spectral distributions.**

**Manipulation checks and individual differences.**

**Predictive modeling and brain–behavior.**

**Full fixed-effects table.**

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

| Model                | Predictor                       | OR (95% CI)          | p      |
|----------------------|---------------------------------|----------------------|--------|
| Baseline (high grip) | Intercept                       | 1.98 (1.60, 2.45)    | < .001 |
| Baseline (high grip) | Modality (visual vs. auditory)  | 1.70 (1.48, 1.96)    | < .001 |
| Baseline (high grip) | Stimulus difficulty (linear)    | 4.76 (3.86, 5.87)    | < .001 |
| Baseline (high grip) | Stimulus difficulty (quadratic) | 30.19 (24.85, 36.68) | < .001 |

Table 3: Complete fixed-effects tables for baseline and DSP-augmented models (high-grip trials).

| Model                     | Predictor                       | OR (95% CI)          | p      |
|---------------------------|---------------------------------|----------------------|--------|
| Baseline (high grip)      | Stimulus difficulty (cubic)     | 0.16 (0.14, 0.19)    | < .001 |
| Baseline (high grip)      | Stimulus difficulty (4th order) | 1.75 (1.54, 2.00)    | < .001 |
| DSP-augmented (high grip) | Intercept                       | 1.99 (1.61, 2.46)    | < .001 |
| DSP-augmented (high grip) | Modality (visual vs. auditory)  | 1.70 (1.47, 1.96)    | < .001 |
| DSP-augmented (high grip) | Stimulus difficulty (linear)    | 4.76 (3.86, 5.87)    | < .001 |
| DSP-augmented (high grip) | Stimulus difficulty (quadratic) | 30.19 (24.85, 36.68) | < .001 |
| DSP-augmented (high grip) | Stimulus difficulty (cubic)     | 0.16 (0.14, 0.19)    | < .001 |
| DSP-augmented (high grip) | Stimulus difficulty (4th order) | 1.75 (1.54, 2.00)    | < .001 |
| DSP-augmented (high grip) | Tracking power 0.5–3 Hz (z)     | 1.00 (0.93, 1.08)    | 0.965  |
| DSP-augmented (high grip) | Tremor power 8–12 Hz (z)        | 0.96 (0.88, 1.04)    | 0.282  |

## 5 Reproducibility

Analysis notebooks live in `analysis/`:

| Notebook                                  | Purpose                                                  |
|-------------------------------------------|----------------------------------------------------------|
| <code>01_ingest.qmd</code>                | Trial table construction and spectral feature extraction |
| <code>02_cognitive_motor_model.qmd</code> | Mixed-effects models and LC brain–behavior analyses      |
| <code>03_hero_visuals.qmd</code>          | Figure generation for manuscript and portfolio           |

Render from the repository root:

```
quarto render analysis/01_ingest.qmd
quarto render analysis/02_cognitive_motor_model.qmd
quarto render analysis/03_hero_visuals.qmd
```

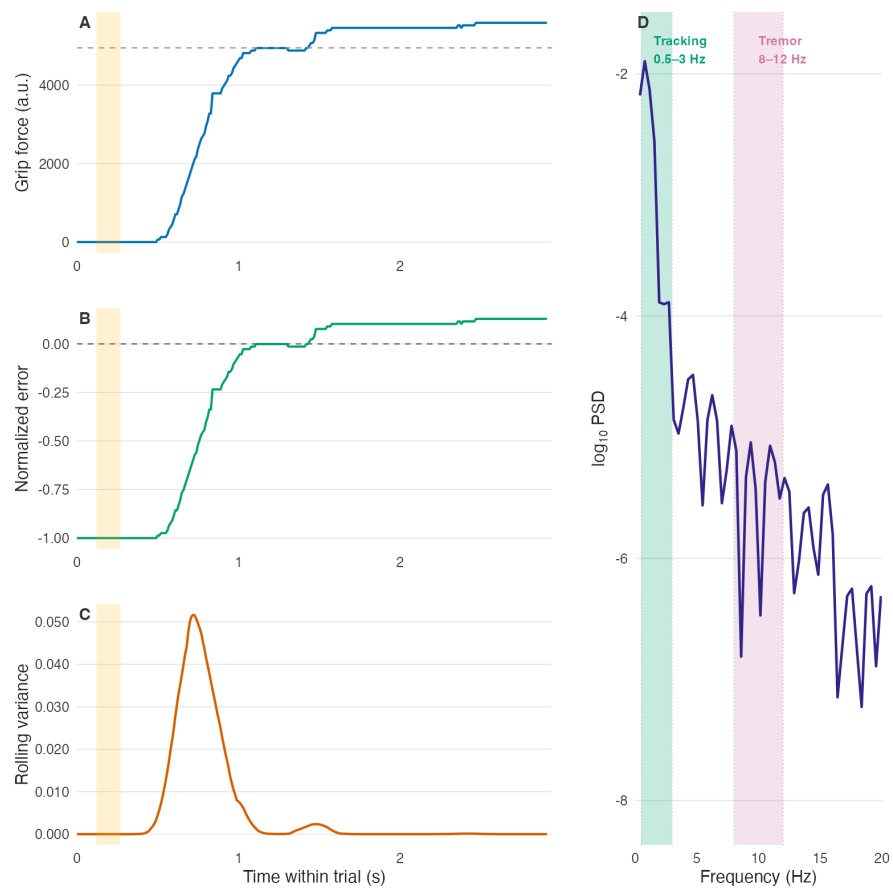


Figure 1: Single-trial signal chain from grip force to spectral phenotype.

### The Cognitive Shock: Probe-Locked Motor Spectrogram

BAP171 · vis · session 2 · trial 8 · incorrect response

A = normalized error; B = short-time spectrum. Vertical gold line = oddball probe ( $t = 0$ ). Grey shading = pre-probe baseline. Green band = tracking (0.5-

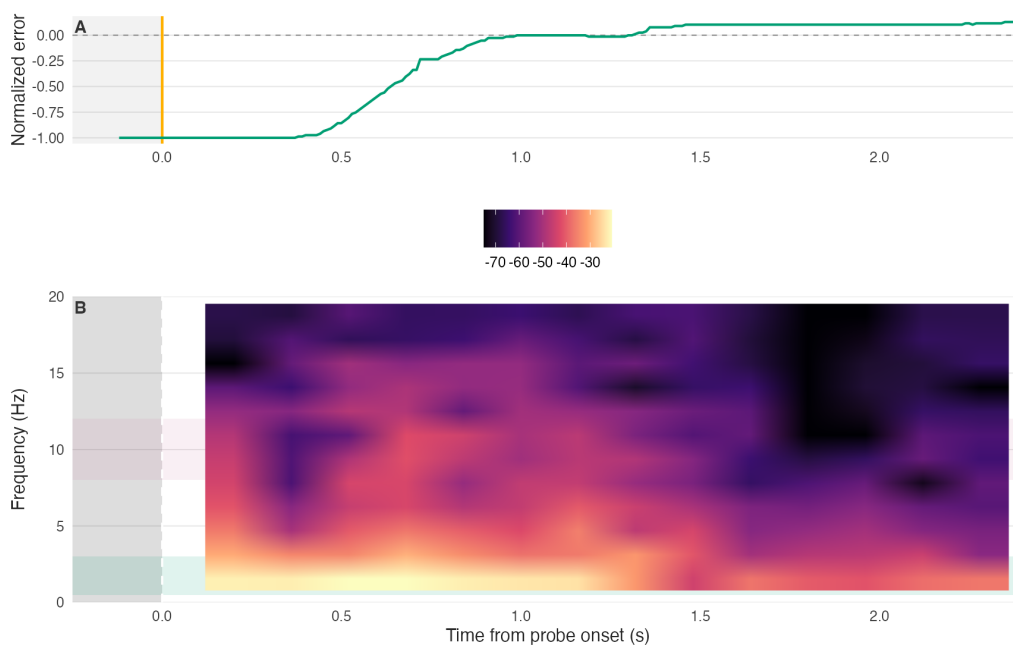


Figure 2: Probe-locked motor spectrogram aligned to oddball onset.

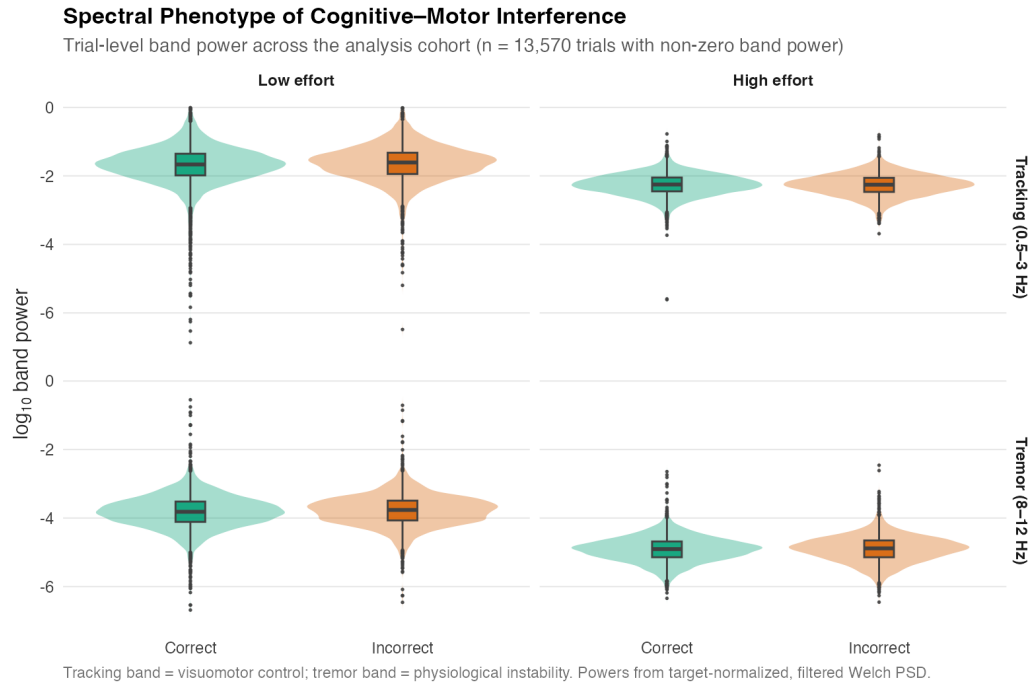


Figure 3: Trial-level log band-power by grip effort and perceptual outcome.

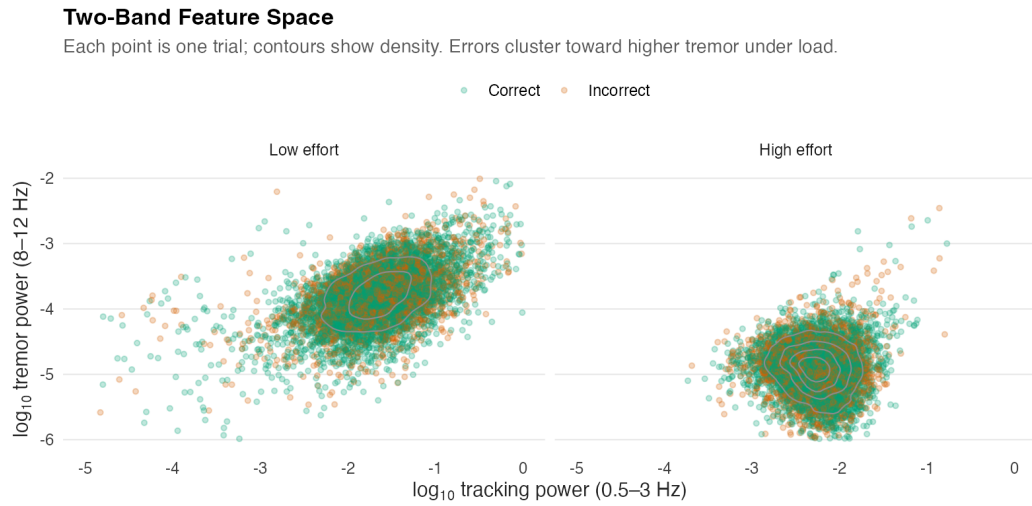


Figure 4: Two-band feature space (tracking vs. tremor), split by grip effort.

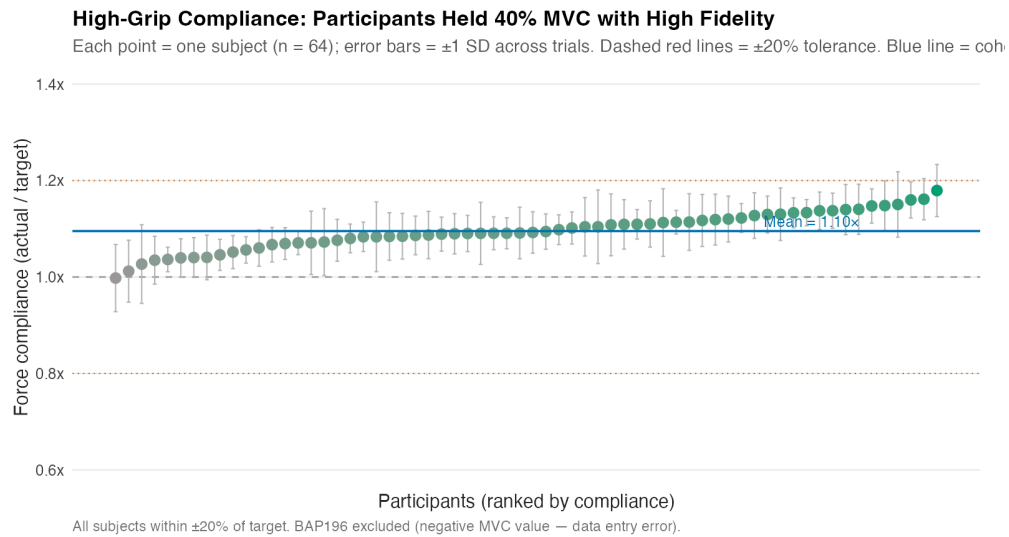


Figure 5: Per-subject high-grip force compliance (actual / target).

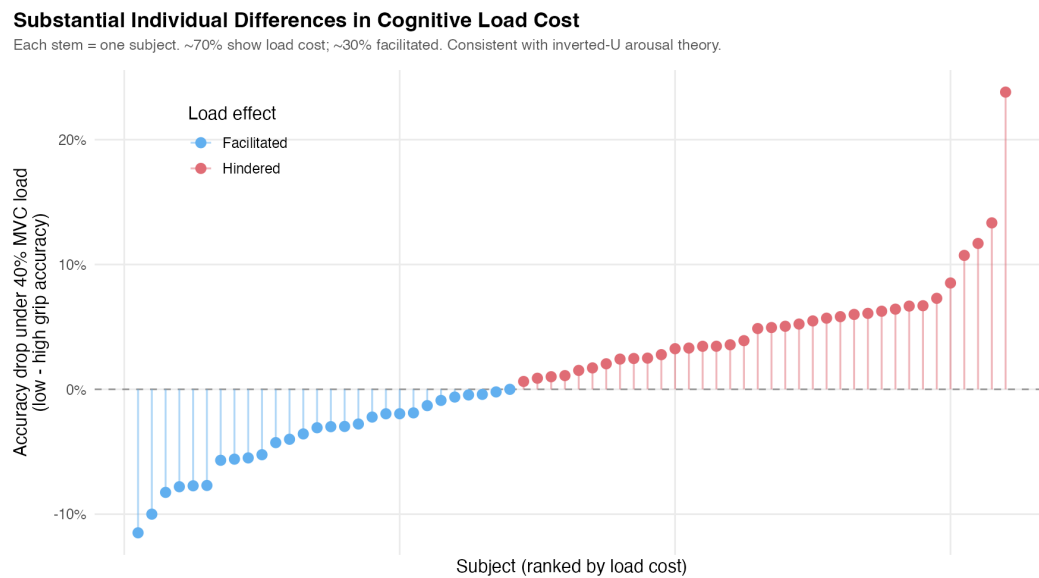


Figure 6: Subject-level accuracy drop under high versus low grip (caterpillar plot).

### Physical Load Degrades Accuracy Without Buying Speed

Each line = one subject. Diamond = grand mean  $\pm$  SE. High grip shifts cloud **down** (lower accuracy) with no systematic RT change.

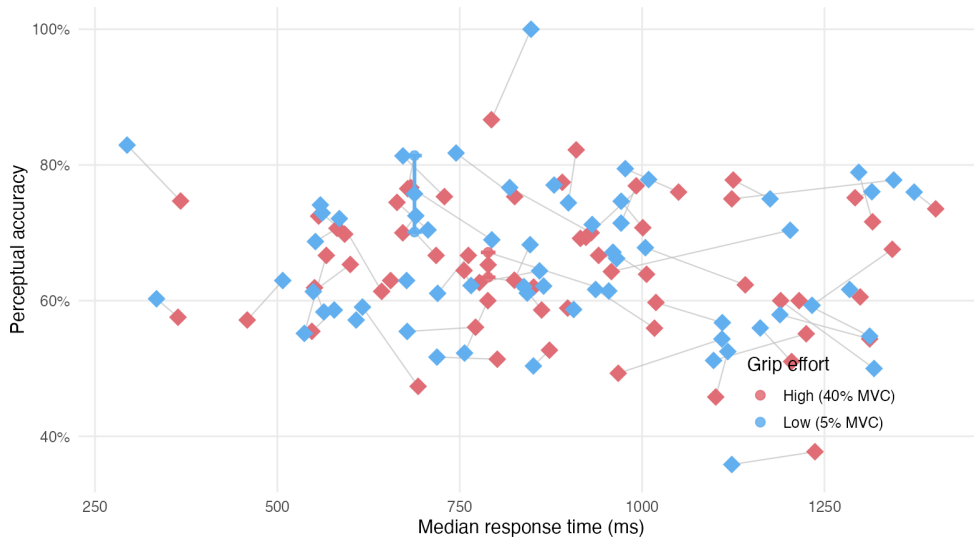


Figure 7: Speed-accuracy trade-off under low versus high grip.

### Cross-validated prediction of perceptual accuracy

Random forest, subject-wise cross-validation, high-grip trials only

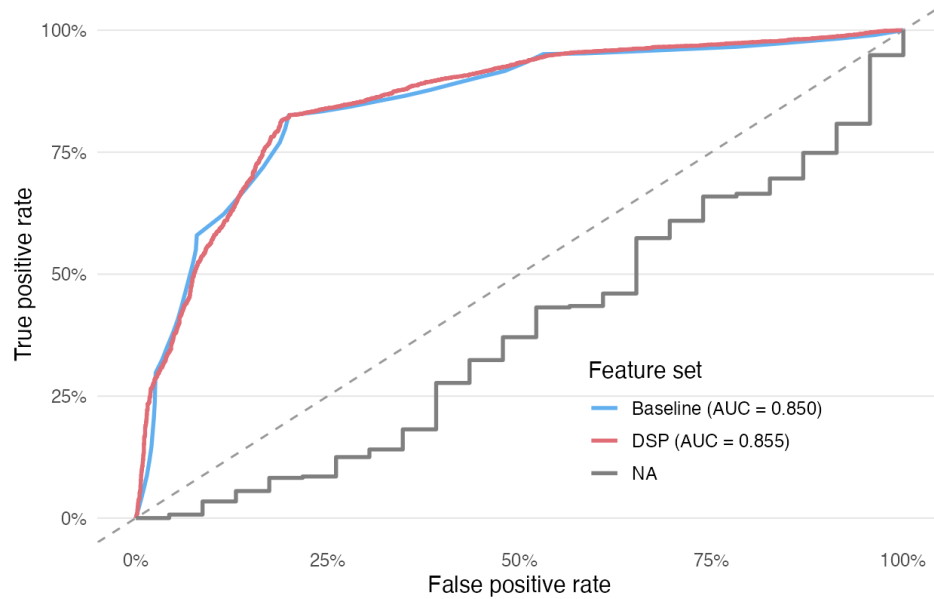


Figure 8: Cross-validated ROC curves (random forest, high-grip trials).

Exploratory LC microstructure and load-related motor reactivity  
N = 55 with LC metrics and grip-force features

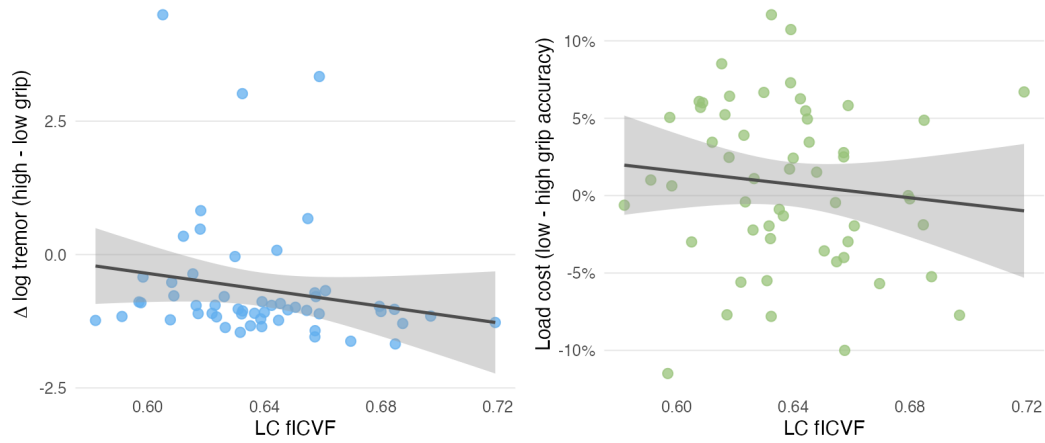


Figure 9: Exploratory LC fICVF associations with load-related tremor reactivity and perceptual load cost.

```
quarto render manuscript/manuscript.qmd
quarto render manuscript/manuscript-biorxiv.qmd --to biorxiv-typst
quarto render manuscript/supplementary.qmd
```

Local data paths are configured in `config/paths.local.yaml` (not tracked in git).